

24PY901
Quantum Principles, Structures and Computing
Module 1 – Unit 1

Course instructor: Dr. Sreekar Guddeti

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Solutions to Assignment 2: Physics of Smartphone

Sensors, Communication, Interface, Internals, and Energy

References:

1. Avadhanulu, Chapters 24. Quantum Hypothesis, 26. Matter waves, 27. Quantum mechanics
2. World Wide Web

Instructions:

- Choose the **most appropriate answer** for each question.
- Some technologies may involve more than one physics concept.
- Use this questionnaire to relate physics concepts to real-world devices you use daily.

Sensors

1. A smartphone camera sensor converts incident light into electrical signals. Which physics concept best explains this? (a) Photoelectric effect

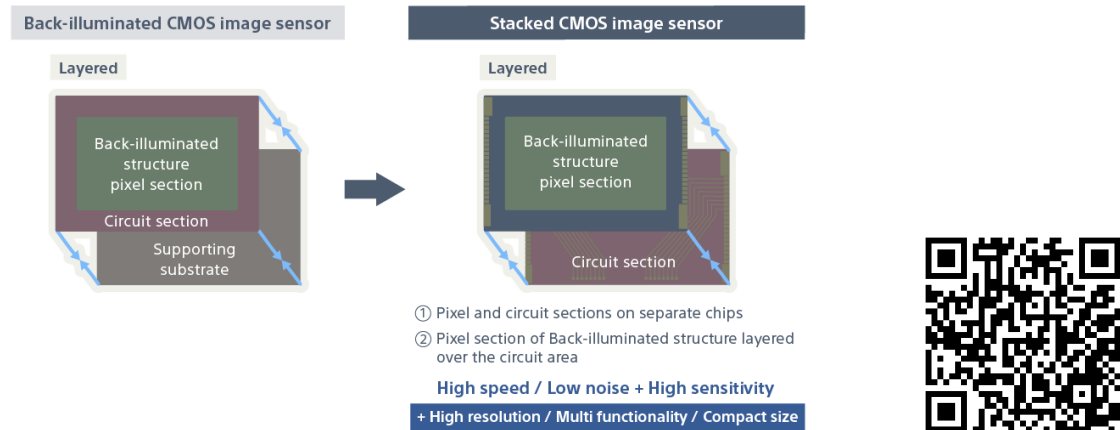


Figure 1: White paper on stacked CMOS image sensor from Sony.

2. Touchscreens detect finger position by sensing changes in electric fields. This technology primarily relies on: (a) Capacitance

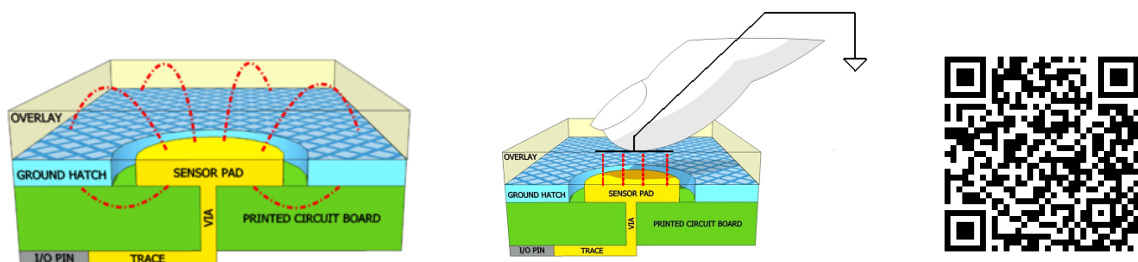


Figure 2: White paper on touch sensor from Infineon.

3. Automatic screen rotation in smartphones is achieved using (a) Accelerometer (Newton's laws), and (b) Gyroscope (Angular momentum)



Figure 3: White paper on (a) accelerometer and (b) gyroscope from Analog Devices.

4. Fingerprint sensors in smartphones operate using: (a) Optical reflection, (b) Electrostatics, and (c) Semiconductor physics

5. Face recognition systems often use infrared depth sensing. The underlying principle involved is: (b) Time-of-flight of photons

Communication

6. Mobile phones communicate with base stations using: (a) Radio / microwave electromagnetic waves
7. Mobile signals can bend around buildings due to: (a) Wave nature of electromagnetic radiation
8. Noise-cancelling headphones work by producing sound waves of opposite phase. This phenomenon is known as: (b) Destructive interference

Interface

9. Light emission in LED and OLED displays occurs due to: (b) Electroluminescence
10. Change in appearance of phone screens at large viewing angles is mainly due to: (a) Polarization effects
11. Blue-light filters in smartphones reduce eye strain by modifying: (a) Wavelength range

Internals

12. Proximity sensors in smartphones switch off the display during calls using: (a) Reflection of infrared radiation
13. The functioning of semiconductor chips in mobile phones is based on: (a) Energy band gap
14. Performance degradation of smartphones at high temperatures is due to: (a) Increased lattice vibrations, (b) Reduced carrier mobility, and (c) Increased thermal noise

Energy

15. Heat generation during fast charging of mobile phones is governed by: (a) Joule heating (I^2R loss)
16. Wireless charging of smartphones works on the principle of: (a) Electromagnetic induction

Concept Mapping

17. Match the following technologies with the most relevant physics concept:

Technology	Physics Concept
Camera sensor	Photo-electric effect
Touchscreen	Electrostatics
Speaker	Electromagnetic induction
Microphone	Electromagnetic induction
GPS	Time of flight measurement of EM waves
Battery	Electrochemistry

18. Choose any one feature of a smartphone and explain the physics behind its operation in **3–4 lines**.